

Financial Literacy Among Teacher Candidates: Competence, Calibration and Behavior.*

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June 5, 2026

Despite its relatively recent inclusion as a strand in the 2020 K-8 Ontario Mathematics Curriculum, financial literacy remains underexamined within teacher education research, particularly in relation to how educators conceptualize and develop capacity to teach it. Indeed, research on financial literacy rarely examines teachers' own financial literacy, even less how it relates to their personal financial behaviors or their perceived self-efficacy to teach these topics. This study conceptualizes readiness to teach financial literacy among teacher candidates as a multidimensional construct shaped by the interaction between (1) financial literacy knowledge and behaviour, (2) teaching self-efficacy, as well as (3) the calibration between perceived and demonstrated competence. Drawing on survey data from 145 pre-service teachers in Ontario, we examine the relationship between teacher candidates' content knowledge, teaching self-efficacy, and capacity to interpret and teach financial situations. Results...

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*Code and data are available at: https://github.com/mejiama16/Financial_Literacy_Study.

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1 Introduction

Financial literacy is broadly defined as having the knowledge, skills and confidence to make responsible financial decisions (Nicolini et al., 2013; OECD, 2017). It is conceptualized by the OECD as dependent on an individuals’ financial knowledge, financial behaviours, and financial attitudes. Financial knowledge refers to the basic knowledge of financial concepts and the associated numeracy skills needed to apply their knowledge in financial contexts (OECD, 2023). It includes understanding financial concepts such as inflation, interests on loans, risk and return, the time value of money, risk diversification, and simple and compound interest. It also encompasses the numeracy skills needed to apply these skills, for instance, by computing simple and compound interest calculations. Financial behaviour refers to the behaviours and actions that affect an individual’s financial situation and well-being in the short and the long term (OECD, 2023). Individuals with positive financial behaviours keep track of their expenses and personal finances, plan for saving and investing in the short and long term, and make decisions relating to financial products and services by considering multiple options. Last, financial attitudes refers to the extent that attitudes towards money influences financial decisions and behaviours. It is a recognition that even if the individual has the knowledge and ability, this might not align with their actions due to their attitudes.

Existing studies have shown that financial illiteracy is widespread, with many individuals lacking even basic economic knowledge (Lusardi & Mitchell, 2007, 2011). In 2022, thirty-nine countries assessed their citizens’ financial literacy levels through the OECD/INFE Toolkit for Measuring Financial Literacy and Financial Inclusion. They found that the average financial literacy score among all participants was 60%, with only 34% of adults reaching the target financial literacy score of 70%. In the Canadian context, cross-country studies have placed Canada above other OECD countries like the U.K. and the U.S. in terms of financial literacy (Nicolini et al., 2013). However, a study measuring financial literacy found that only 42% of Canadians answered basic questions on interest rates, inflation, and risk diversification correctly (Boisclair et al., 2014).

The math educational community has increasingly recognized the importance of teaching financial literacy to students from an early age (OECD, 2005; Compen et al., 2019; Lusardi, 2019) with several countries, including the U.K., Spain, Ireland, Denmark, Belgium, the Czech Republic, and Estonia, responding with initiatives to teach financial literacy to their citizens through mandatory financial literacy curricula in schools (Compen et al., 2019). By teaching students about healthy spending, effective saving strategies, retirement-planning, and beyond, the integration of personal finance education is hypothesized to be a proactive tool to support individuals in managing their finances. In Ontario, personal finance is taught in some high school mathematics courses in Grades 11 and 12, particularly those in the college and work-place streams. However, given citizens’ changing economic realities, the Ministry of Education has expanded educational initiatives to enhance financial literacy education by introducing a financial literacy strand to their K-8 Mathematics Curriculum (2020) and the Grade 9 Mathematics Curriculum (2021), a strand designed to equip students with the skills and knowledge to “manage their personal financial well-being with confidence, competence, a critical and compassionate awareness of the world around them” (Ministry of Education, 2020). For instance, students in Grade 9 learn to “Identify a past or current financial situation and explain how it can inform financial decisions by applying an understanding of the context of the situation and related mathematical knowledge” (Ministry of Education, 2021). They also learn how to modify budgets according to different changes in circumstances, how to compare the effects of different interest rates and about appreciation and depreciation.

Curricular inclusion of financial literacy concepts creates new expectations for teachers’ content and pedagogical knowledge; that is, their understanding of the subject matter itself and of the teaching methods and learning processes needed to support learners (Shulman, 1986). Consequently, financial literacy must be positioned as a pedagogical and curricular concern for mathematics teacher education. Yet, despite their importance in educating the next generations, research on financial literacy rarely examines teachers’ own financial literacy conceptual knowledge, even less how it relates to their personal financial behaviors (Compen et al., 2021) and pedagogical knowledge. Existing literature has explored teachers’ perceptions of inflation (Bansilal, 2017), attitudes toward financial literacy education (Henderson et al., 2021), and self-assessed competence (De Beckker et al., 2019), finding that teachers’ financial literacy competencies and confidence remain suboptimal across different contexts and evaluations, highlighting the need for additional support to strengthen both their content knowledge and pedagogical readiness.

Beyond content and pedagogical knowledge of financial literacy, it is essential to consider teacher candidates’ self efficacy, as this construct shapes teachers’ confidence and can influence how meaningfully they engage with financially contextualized mathematical tasks in their classrooms. Self-efficacy has been recognized as a key factor shaping teachers’ instructional strategies and even student achievement, with research showing that teachers with higher teaching self-efficacy are more likely to employ student-centered teaching practices, supporting deeper conceptual understanding among students (Stipek et al., 2001). Self-efficacy research in mathematics preservice teacher education also highlights the role of training and experience in shaping perceived capability (Fitzmaurice et al., 2014; Twohill et al., 2023).

Specifically in the context of mathematics education, mathematics teacher self-efficacy can be further conceptualized into two main components: mathematical subject knowledge itself, referred to as mathematics self-efficacy, and pedagogical knowledge, referred to as mathematics teaching self-efficacy. Teaching self-efficacy is an important attribute to promote and develop among pre-service teachers since it both positively impacts personal academic outcomes and future teaching prospects (Segarra & Julià, 2022; Charitaki et al., 2025). For example, higher teacher self-efficacy leads to improved personal academic performance in pre-service teachers (Segarra & Julià, 2022). Additionally, teachers' beliefs of their self efficacy better predicts their sense of responsibility for student achievement than other variables like mathematics teaching outcome expectancy (Charitaki et al., 2025).

Specifically, teaching efficacy in financial literacy is an even more understudied topic in teacher self-efficacy research despite teachers' pedagogical knowledge of financial literacy instruction becoming increasingly relevant with the increasing adoption and implementation of financial literacy curriculum in Mathematics (Ministry of Education, 2020; Financial Consumer Agency of Canada, 2025). Many studies find that teachers are less confident in teaching financial literacy (Sawatzki & Sullivan, 2017; Henderson et al., 2021; Wait, 2024). For example, self-reported financial literacy teaching self efficacy was lower than general teaching efficacy in an Australian teacher sample (Wait, 2024). Sawatzki & Sullivan (2017) also found a discrepancy between the number of teachers who perceived themselves as financially literate (more than 75%) and the number of teachers who felt confident teaching financial literacy (around 50%). This suggests that conceptual and pedagogical knowledge around financial literacy should not be assumed to transfer automatically from general mathematics self-efficacy. Instead, financial literacy demands specific attention. Taken together, this literature signals that readiness to teach financial literacy among pre-service teachers must be conceptualized as a multi-dimensional construct that considers the interplay between financial knowledge and behaviours, pedagogical self-efficacy, and alignment between perceived and demonstrated competence.

2 Methodology

The study used a cross-sectional correlational design, drawing on Microsoft survey responses examining financial literacy readiness among teacher candidates. The 145 participants of this study were teacher candidates at a graduate institution in Ontario (109 female, 35 males, 1 non-binary). More specifically, the sample comprised two out of the three teaching divisions: 67 Primary/Junior (P/J; Kindergarten to Grade 6, $N = 67$) and 78 Junior/Intermediate (J/I; Grades 4 to 10, $N = 78$). There were no participants from the Intermediate/Senior division (I/S; Grades 7 to 12). Ethics Approval was obtained from the institution where this study took place and consent was collected from all participants completing the survey; the survey data was obtained from teacher candidates in their mathematics pedagogy course.

3 Instrument

The survey used in this study was composed of four sections designed to capture demographic characteristics, teaching self-efficacy, financial literacy knowledge, and financial behaviour. Together, these measures were intended to provide a multidimensional view of teacher candidates' preparedness to engage with financial literacy as part of their mathematics teaching.

1) Demographic Questions: participants reported demographic information including age, gender, country of origin, and first language.

2) Mathematics and Financial Literacy Teaching Self-Efficacy: Teaching self-efficacy was measured using items adapted from the Personal Mathematics Teaching Efficacy subscale of the Mathematics Teaching Efficacy Beliefs Instrument (MTEBI; Enochs et al., 2000). This measure was chosen because it has been widely used in mathematics teacher education research and due to its focus on teachers' perceived capability to teach subject-specific content, thereby allowing substitution to align with the study's focus on financial literacy. The original subscale was shortened to four items due to survey length constraints. The four items were selected based on the directness in measuring self-efficacy and mostly excluded items that also referred to other constructs such as anxiety. Item responses were measured on a 5-point Likert scale (1 = strongly disagree and 5 = strongly agree). This adapted scale was used to measure mathematics teaching self-efficacy (MTSE). An equivalent scale with identical question formulation was adapted to measure financial literacy teaching self-efficacy (FLTSE).

3) Financial Literacy Knowledge: Conceptual competency was measured through nine multiple-choice questions: 4 multiple-choice questions were used to assess basic financial literacy knowledge, based on content in inflation, interest rates, time value of money, and numeracy, and 5 multiple-choice questions measured advanced financial literacy knowledge, based on content in stocks, bonds, and mutual funds. Each question was dichotomously scored (1 = correct, 0 = incorrect). Basic (range: 0-4) and advanced scores (range: 0-5) were summed to form the total competency score (range: 0-9). The questions measuring both basic and advanced financial literacy were constructed upon van Roij, Lusardi & Alessie's (2011) instrument to evaluate financial literacy. A subset of questions from this paper were selected for their clear distinction between basic and advanced financial literacy, their cross-country adaptability, and their established link to stock market participation outcomes. Box 1 and 2 in the Appendix contain the selected questions that were used in our survey.

4) Financial Behaviour: Participants also reported whether they participate in the stock market (i.e. hold individual stocks, bonds or mutual funds), and the platforms and sources they use for financial information.

4 Results

4.1 Financial Literacy Knowledge

4.2 Mathematics and Financial Literacy Teaching Self Efficacy

4.3 Calibration

5 Limitations

Interpretations of these findings should be made with caution, as several limitations constrain this study. First, this study used convenience sampling and drew on a single regional group of teacher candidates, which limits the generalizability of the findings to broader populations. Moreover, the data is self-reported, which introduces the possibility of inaccurate or socially desirable responses, particularly regarding stock market participation. We should also consider that the survey was relatively long, and it is possible that some answers might have been affected by survey fatigue. Lastly, the sample is relatively small for the ideal statistical significance and may not be representative of all teacher candidates. It is also worth noting that previous research has shown that increased financial knowledge does not always lead to improved financial behaviours or outcomes (Fernandes et al., 2014). In relation to self-efficacy analysis, conceptual competency was assessed using nine multiple-choice questions, which may not fully reflect the depth of financial literacy knowledge needed for teaching. Similarly, financial literacy teaching self-efficacy was measured using a four-item scale, which may not fully reflect the complexity of perceived teaching capability. In addition, the self-report measure is susceptible to social desirability bias, indicating how responses may be influenced by individuals' tendency to present themselves in a favorable way (Paulhus, 1991).

6 Discussion

Teacher candidates are an interesting group to study because they typically anticipate stable employment and a defined retirement plan through the Ontario Teacher's Pension plan. As proposed by Buckland (2010), people's financial literacy needs vary across socioeconomic groups. For example, since low-income Canadians do not save as much for retirement compared to middle-income and high-income, this type of financial literacy is less relevant for them than for example, knowledge about income security programs (Buckland, 2010). Similarly, future teachers may not feel the same urgency to participate in the stock market or stay informed about the latest financial tools, given their stable career prospects and guaranteed pension. This raises the question of whether this stability shapes their financial literacy and their likelihood to engage in financial behaviors such as investing. Our results indicate that this hypothesis likely doesn't hold, since 42% of teachers in our sample invest in the stock

market, a proportion that is statistically significantly higher than the average in Canada, 33% ($p = 0.011$, $z = 2.30$).

Although FLTSE was conceptualized differently, these results are consistent with existing literature that FLTSE is lower than general teaching self-efficacy in an Australian pre-service teachers sample (Wait, 2024). In this case, general teaching efficacy and MTSE are comparable in the way that they both include broader, well-established and more familiar subjects or topics.

Possible explanations for these results might include the recency of the implementation of FL in the Ontario curriculum. As the FL strand in the K-12 curriculum is relatively new, there might be less opportunities for vicarious learning like teaching demonstrations and hands-on practice with successful experiences to improve their FLTSE compared to well-established topics in mathematics. Additionally, the participant sample likely included pre-service teachers who were not taught FL in local schools in their own elementary school days since the corresponding curriculum was not introduced yet. If there were fewer experiences of being taught FL in a school context, there were likely less opportunities for vicarious observation of FL teaching and potentially lower subject efficacy, which would in turn negatively affect FLTSE.

Moreover, the difference between FLTSE and MTSE suggests that opportunities that improve self-efficacy in teaching mathematics might not be fully generalizable to FL even if it is also part of the mathematics curriculum. It might therefore be beneficial to create learning opportunities specific to FL so that they are not sidetracked by other mathematical topics. Future research should replicate this study with larger and more diverse samples of teacher candidates across different provinces and career stages, incorporating measures of socioeconomic status to better account for its influence on both financial literacy and investment behavior. Longitudinal designs tracking teacher candidates from pre-service through early career would shed light on whether financial literacy and stock market participation evolve as teachers transition into stable employment and become able to invest.

The study examined whether financial literacy teaching self-efficacy aligns with conceptual competency among pre-service teachers and whether miscalibration patterns exist across competency quartiles. Overall, the positive association between teaching self-efficacy and competency indicates that participants who feel more capable of teaching financial literacy generally demonstrate higher conceptual knowledge. This finding is consistent with research in mathematics education suggesting that knowledge and self-efficacy can be related and may need to be developed in coordination with preparation for teaching (Alshehri & Youssef, 2022).

The most notable finding of this study is the miscalibration pattern, where participants in the lowest competency quartile (Q1) tended to overestimate their teaching ability (positive miscalibration mean: $SE_z > Comp_z$), while those in the highest competency quartile (Q4) showed underestimation (negative miscalibration mean: $SE_z < Comp_z$). In the Dunning-Kruger literature, this kind of pattern is interpreted as lower performers overestimating and higher performers underestimating their own performance, and this effect occurs because people need a minimum level of knowledge to assess their own ability accurately (Kruger & Dunning, 1999).

In the context of this study, since pre-service teachers in the lowest competency quartile have not seen the full complexity of financial literacy, they assume that teaching it is easier than it actually is. In contrast, pre-service teachers in the highest competency quartile, with a deeper understanding, are more aware of what they do not know, leading them to underestimate their teaching capability.

The additional findings provide further insights into how self-efficacy and competency are shaped by financial engagement and teaching context. First, pre-service teachers who currently invest showed higher self-efficacy, suggesting that personal financial experience may help them feel more capable of teaching financial concepts. This interpretation is consistent with an idea from mathematical research that self-efficacy is shaped by experiences that create a sense of familiarity (Charalambous et al., 2008). Second, the intention to engage in future investment was associated with higher conceptual competency. Without claiming directionality, this pattern could be explained by the fact that pre-service teachers interested in future participation in financial markets may pay more attention to or actively seek financial information or knowledge. Third, pre-service teachers in the J/I division reported higher self-efficacy than those in the P/J division. This may be explained by differences in curriculum demand: financial literacy is included in Grade 9 mathematics, making it more salient for those preparing to teach upper-level students.

7 Conclusion

While financial literacy has been increasingly embedded within mathematics curricula, there remains limited understanding of teacher candidates' knowledge, behaviors, and pedagogical capacity required to teach these concepts effectively. This study conceptualized readiness to teach financial literacy by looking at the interplay between financial knowledge and behaviours, pedagogical self-efficacy, and alignment between perceived and demonstrated competence. Our findings suggest that...

By gaining a better understanding of the conditions that support meaningful financial literacy instruction in mathematics classrooms, ...

Teacher candidates in this sample showed moderate overall financial literacy, with stronger performance on basic concepts than advanced. Additionally, 42% of teacher candidates invest, notably higher than the roughly one-third of Canadians nationally, which challenges the expectation raised in the introduction that stable employment and a guaranteed pension might reduce urgency to invest. While regression models did not yield statistically significant predictors, descriptive results suggest a positive relationship between financial literacy and stock market participation, in line with van Rooij et al. (2011).

Specific areas of improvement found by the study include interest rates and fixed income, which highlighted a mismatch between procedural and conceptual knowledge of financial concepts. These findings, combined with the unprecedented rates at which younger generations

are investing given world uncertainty, underscore the importance of strengthening advanced financial knowledge in teacher preparation programs, with attention not only to procedural calculation skills but to the conceptual understanding that enables richer and more equitable curriculum delivery.

8 Assumptions

For the multiple linear regression, observations are assumed independent given that each participant completed the survey individually. It is important to note that the outcome variable (financial literacy index) is discrete rather than truly continuous; for example, the basic index takes up to five possible values given there are four questions. Nevertheless, linear regression is robust to mild violations of continuity (Schmidt & Finan, 2018) and represents the conventional approach in the financial literacy literature (van Rooij et al., 2011; Nicolini et al., 2013). Hence, this method will be employed using the overall index as a dependent variable to better approximate continuity. Following model fitting, residual plots confirmed linearity and homoscedasticity, VIF values were below 5 indicating no multicollinearity, and the Q-Q plot revealed only a mild deviation from normality at the tails. For the logistic regression, the binary outcome variable (invest/not invest) satisfies the core requirement of the model by design. No multicollinearity was detected ($\max \text{VIF} = 2.8$), and the Box-Tidwell test supported the linearity of log-odds for all continuous predictors. However, with 61 investors against a recommended minimum of 100 for a 10-predictor model (Peduzzi et al., 1996), the sample size adequacy threshold was not fully met, implying that findings are reported as exploratory and should be interpreted with caution, pending replication with a larger, more diverse sample.

9 References